

**AI Project**

**BSAI**

# **Emotion Detection from Text Messages**

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# Project Overview

- ❑ This project is a text-based emotion detection system
- ❑ It identifies emotions from user input text
- ❑ Built using rule-based Natural Language Processing (NLP)
- ❑ Emotions include:
  - a. Happy
  - b. Sad
  - c. Angry
  - d. Fearful
  - e. Excited
  - f. Neutral

# Objective

- ☐ Detect emotions from user text messages
- ☐ Provide supportive responses based on emotion
- ☐ Store user emotional history
- ☐ Analyze emotional patterns over time
- ☐ Generate emotional warnings if needed

# Libraries Used

- ❑ **os** → File and folder management
- ❑ **datetime** → Timestamp logging
- ❑ **pandas** → Dataset processing (CSV file handling)
- ❑ **These libraries support file handling, data analysis, and time tracking.**

# System Workflow

- ☐ User enters text
- ☐ Text is cleaned (stop words + punctuation removed)
- ☐ Emotion detection is performed
- ☐ Emotion is classified:
  - a. Phrase matching
  - b. Strong word matching
  - c. Dataset keyword mapping
- ☐ Result is displayed + stored in file

# Text Cleaning Process

- ☐ Converts text to lowercase
  - ☐ Removes punctuation
  - ☐ Splits text into words
  - ☐ Removes stop words
  - ☐ Returns clean word list
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- ☐ Improves accuracy of emotion detection
  - ☐ Removes unnecessary words

# Emotion Detection Techniques

## ☐ Phrase Matching

- a. e.g. “very happy”, “not feeling well”

## ☐ Strong Word Matching

- a. happy → amazing, success
- b. sad → depressed, hopeless

## ☐ Dataset Keyword Mapping

- a. Uses CSV file (keyword → emotion)

## ☐ Negation Handling

- a. “not happy” → Sad instead of Happy

# Scoring Mechanism

- ❑ Each emotion is assigned a score
- ❑ Weights:
  - a. Strong words → high weight
  - b. Dataset keywords → medium weight
  - c. Negation modifies meaning
- ❑ Final emotion = highest score
- ❑ If confidence is low → Neutral



# User Management System

- ☐ Each user has:

- a. Username
- b. Age
- c. Personal file stored in “users” folder

- ☐ Features:

- a. Save emotional history
- b. Load past records
- c. Timestamp tracking

- ☐ Enables personalized emotion tracking

# Thank You